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**School of Computer Science Engineering and Information Systems**
**Fall Semester 2026-2027**

**Department of Information Technology**
**BITE497J - Project - I**
**Review – 1**

Date: 19.08.2026

# CogniShield: Edge-Based Adversarial Immunization for Digital Media

**23BIT0030 – Rohit Kumar**
**23BIT0067 – Shivansh Rai**
**23BIT0241 – Aviral Bhan**

Under the Guidance of
**Dr. Angulakshmi M**

**SCORE**

**Guide Signature with date**

Guide Name: Dr. Angulakshmi M

**Internal Examiner -1**

Dr. Sudha M

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**Internal Examiner-2**

Dr. Vivekananda G N

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## ABSTRACT:

CogniShield is an edge-based adversarial immunization system designed to protect digital images from unauthorized AI-based manipulation. With the rapid growth of generative AI and deepfake tools, personal images can be misused to create fake or manipulated media. Most existing systems focus on detecting fake content after it is generated, but CogniShield follows a proactive approach by protecting the image before it is uploaded or shared.

The proposed system adds small, low-visibility perturbations to an input image. These perturbations are designed to disturb AI image-to-image manipulation models while preserving human visual quality. The system evaluates the protected image using image quality metrics such as PSNR, SSIM, MSE, and pixel difference values. It also tests robustness against JPEG compression, since online platforms often compress uploaded images. Additionally, CogniShield includes a cryptographic provenance module that signs the protected image using SHA-256 hashing and Ed25519 digital signature verification. This helps verify whether the image has been modified after protection.
The prototype is implemented using Python, Streamlit, Pillow, NumPy, scikit-image, cryptography, PyTorch, and Hugging Face Diffusers. The system works locally, supporting privacy-preserving edge computation. CogniShield demonstrates a practical step toward proactive digital identity protection against generative AI misuse.



Keywords:
Generative AI Security, Adversarial Perturbation, Deepfake Prevention, Digital Media Protection, Cryptographic Provenance, Edge Computing, Image Authentication, JPEG Compression Robustness, Privacy-Preserving Technology.


## [1.] INTRODUCTION:

The development of generative artificial intelligence has made image editing, face manipulation, and deepfake generation easier than ever before. Modern AI models can transform images, modify faces, generate realistic portraits, and create manipulated media with very little technical effort. While these technologies have useful applications in entertainment, design, education, and research, they also create serious risks for privacy, digital identity, and cybersecurity.

Traditional media security systems mainly focus on detecting whether an image or video is fake after it has already been created. However, this reactive approach has limitations. Once fake media is generated and shared online, the damage may already be done. The victim may face reputation loss, identity misuse, misinformation, or harassment. Therefore, there is a need for a proactive method that protects digital media before misuse happens.

CogniShield addresses this problem by applying adversarial immunization to images. The system adds small perturbations that are difficult for humans to notice but can interfere with AI-based image manipulation models. Along with image protection, CogniShield also uses cryptographic provenance to verify whether the protected image has been modified after signing. The system is designed to run locally on the

user's device, reducing privacy risks caused by uploading personal images to cloud-based protection services.

## [2.] PROBLEM STATEMENT:

Generative AI models and deepfake tools can manipulate personal images with high realism, creating risks such as identity misuse, fake media generation, misinformation, and privacy violation. Existing solutions mainly detect fake media after it is created, which is often too late to prevent harm. Many systems also require users to upload images to external servers, creating additional privacy concerns.

Therefore, the problem is to design a local, privacy-preserving system that can protect images before they are shared online by adding low-visibility adversarial perturbations, testing the visual quality trade-off, checking robustness against compression, and verifying image integrity through cryptographic provenance.

## [3.] OBJECTIVES:

- To develop a local image protection system that adds low-visibility adversarial perturbations to digital images.
- To reduce the effectiveness of AI-based image manipulation or image-to-image generation models on protected images.
- To preserve the visual quality of protected images using measurable image quality metrics such as PSNR, SSIM, MSE, and pixel difference.
- To test whether the protected image remains visually acceptable after JPEG compression.
- To implement cryptographic provenance using SHA-256 hashing and Ed25519 digital signatures.
- To build a simple user interface for uploading images, generating protected images, viewing metrics, and downloading results.
- To design the system as an edge-based prototype that runs locally and supports privacy-preserving digital media protection.

## [4.] SCOPE OF THE PROJECT:

The scope of this project is limited to image-based digital media protection. The current prototype focuses on protecting static images from AI-based manipulation. The system accepts image files such as JPG, PNG, JPEG, and WEBP, applies adversarial-style perturbation, evaluates image quality, performs JPEG compression testing, and generates provenance information.

The project includes a Streamlit-based user interface and a command-line pipeline for batch experiments. The optional target model testing module uses SD-Turbo image-to-image generation to compare the behavior of original and protected images.

The project does not currently cover video deepfake protection, audio deepfake protection, real-time mobile deployment, or complete defense against all generative AI models. These can be considered as future enhancements.

## [5.] PROPOSED SYSTEM:

The proposed system, CogniShield, is an edge-based digital media protection prototype. It protects images locally before they are uploaded or shared. The system contains four major modules: image protection, quality evaluation, compression robustness testing, and cryptographic provenance.

First, the user uploads an image through the Streamlit interface. The image is converted into a standard RGB format and processed locally. The protection module generates small high-frequency perturbations and applies them to the image within a fixed perturbation budget. The perturbation is controlled so that the image remains visually similar to the original.

Second, the system evaluates the quality of the protected image. It calculates PSNR, SSIM, MSE, mean absolute pixel difference, and maximum pixel difference. These values help measure the trade-off between protection strength and visual quality.

Third, the system tests compression robustness. The protected image is passed through JPEG compression at different quality levels such as 95, 75, and 50. The system then recalculates quality metrics to check whether the protected image remains usable after compression.

Fourth, the provenance module creates a SHA-256 hash of the protected image and signs it using Ed25519 cryptography. A provenance JSON file is generated. During verification, if the image has changed, the hash will not match and verification will fail.

The system also includes optional SD-Turbo target model testing. This allows comparison between original and protected image outputs when passed through a generative image-to-image model.

## [6.] LITERATURE SURVEY: (minimum 15 papers)

Existing research shows that adversarial perturbations can disturb AI-based image manipulation models, but robustness remains a major challenge. Many perturbations are weakened by compression, resizing, cropping, denoising, or social media processing. The literature also shows that detection-based approaches are reactive and may fail when new generative models appear.

| S.NO | TITLE | MERITS | DEMERITS |
|---|---|---|---|
| 1 | DiffusionGuard: A Robust Defense Against Malicious Diffusion-based Image Editing (Choi et al., 2025) | Targets unauthorized diffusion-based editing with adversarial noise generated at an early diffusion stage; improves robustness to masks, noise removal and transfer attacks while reducing computational cost. | Focused on diffusion-based editing threats; protection still depends on assumptions about the attack setting and target models. |
| 2 | Verifying Provenance of Digital Media: Security Analysis of C2PA and its Implementation (Golaszewski et al., 2026) | Provides a security analysis of C2PA provenance and identifies concrete implementation/specification weaknesses relevant to authenticity verification. | Analyzes provenance security rather than adversarial image protection; provenance alone cannot prevent AI manipulation. |
| 3 | Robust Universal Adversarial Perturbations (Xu & Singh, 2024) | Develops universal adversarial perturbations designed to remain effective under real-world transformations, including scaling, rotation and JPEG compression. | Evaluated primarily on classification datasets rather than diffusion-based image editing; transfer to CogniShield's target generators requires validation. |
| 4 | MobileNetV4: Universal Models for the Mobile Ecosystem (Qin et al., 2024) | Introduces efficient mobile architectures designed for strong accuracy-efficiency trade-offs across CPUs, DSPs, GPUs and edge accelerators. | It is a general mobile vision architecture paper and does not directly address adversarial perturbation generation or image protection. |
| 5 | Distraction is All You Need: Memory-Efficient Image Immunization against Diffusion-Based Image Editing (Lo et al., 2024) | Presents a memory-efficient semantic attack for image immunization against diffusion-based editing and inpainting, reducing the memory burden of earlier approaches. | Protection is targeted at diffusion editing/inpainting and requires further evaluation under broader transformations and adaptive attacks. |
| 6 | Adversarial Robust Safeguard for Evading Deep Facial Manipulation (Guan et al., 2024) | Introduces a learning-based safeguard that generates protection noise in a single forward pass and explicitly | Targets facial manipulation rather than general image editing; robustness and performance depend on the |

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| | | trains for robustness to common perturbations. | manipulation methods represented during training. |
| 7 | Raising the Cost of Malicious AI-Powered Image Editing (Salman et al., 2023) | Establishes the proactive image-immunization approach using encoder-level and diffusion-level adversarial attacks against malicious diffusion-based editing. | Protection can be computationally expensive and later work demonstrates weaknesses under transformations such as JPEG compression. |
| 8 | Can Protective Perturbation Safeguard Personal Data from Being Exploited by Stable Diffusion? (Zhao et al., 2024) | Evaluates protective perturbations under a practical Stable Diffusion threat model and studies purification attacks that can remove protective noise. | Shows that purification can undermine existing protection methods, highlighting the need for stronger robustness mechanisms. |
| 9 | JPEG Compressed Images Can Bypass Protections Against AI Editing (Sandoval-Segura et al., 2023) | Provides direct evidence that JPEG compression can neutralize imperceptible perturbation-based protections, motivating compression-aware defense design. | Primarily diagnoses the weakness rather than providing a complete end-to-end robust protection method. |
| 10 | Watermark-embedded Adversarial Examples for Copyright Protection against Diffusion Models (Zhu et al., 2024) | Combines adversarial examples with personal watermarks to discourage unauthorized imitation by diffusion models and reports transferability across unknown models. | Focuses strongly on copyright protection and visible watermark outcomes rather than purely imperceptible protection for personal photographs. |
| 11 | Glaze: Protecting Artists from Style Mimicry by Text-to-Image Models (Shan et al., 2023) | Uses barely perceptible style-cloaking perturbations to disrupt unauthorized style mimicry by text-to-image diffusion models and evaluates robustness with large artist studies. | Primarily protects artistic style against training/fine-tuning based mimicry rather than direct pixel-level image editing or inpainting. |
| 12 | DCT-Shield: A Robust Frequency Domain Defense against Malicious Image Editing (Bala et al., 2025) | Moves adversarial protection into the DCT/frequency domain and leverages the JPEG pipeline to improve visual quality and robustness against purification. | The method is specifically designed around frequency-domain/JPEG-aware protection and requires further validation across broader deployment conditions. |
| 13 | How Generalizable are Deepfake Image Detectors? An | Systematically evaluates deepfake-image detector generalization and shows that zero-shot performance can be | It studies detection rather than proactive prevention, so it does not directly protect an image before manipulation. |

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| | Empirical Study (Li et al., 2023) | poor across unseen datasets and synthesis methods. | |
| 14 | UCF: Uncovering Common Features for Generalizable Deepfake Detection (Yan et al., 2023) | Improves deepfake-detection generalization by disentangling common forgery features from method-specific and irrelevant features. | Remains a reactive detection approach and does not prevent the generation or editing of manipulated images. |
| 15 | EfficientViT: Memory Efficient Vision Transformer with Cascaded Group Attention (Liu et al., 2023) | Introduces memory-efficient vision-transformer components and demonstrates strong speed/accuracy trade-offs suitable for efficient vision processing. | It is not an adversarial-protection method and requires adaptation before being used inside CogniShield's protection pipeline. |
| 16 | BlurGuard: A Simple Approach for Robustifying Image Protection Against AI-Powered Editing (Kim et al., 2025) | Improves protection robustness against noise-reversal techniques by applying adaptive per-region Gaussian blur to the protective noise spectrum. | Adds another robustness-processing stage and is primarily evaluated as an enhancement to existing protection methods. |
| 17 | Privacy Protection in Personalized Diffusion Models via Targeted Cross-Attention Adversarial Attack (Xu et al., 2024) | Introduces cross-attention-targeted adversarial protection for personalized diffusion models and aims to protect identity/content with lower perturbation levels. | Focused on personalized diffusion/fine-tuning scenarios; broader transfer to general image-editing pipelines needs additional evaluation. |

### [6.1] FINDINGS IN LITERATURE SURVEY:

- Recent protection research (DiffusionGuard, PhotoGuard, Distraction is All You Need, ARS, DCT-Shield, BlurGuard, Glaze and related adversarial-protection methods) demonstrates that carefully designed perturbations can proactively disrupt unauthorized generative-model use, supporting the core hypothesis behind CogniShield.
- A recurring weakness in the reviewed protection literature is sensitivity to image transformations and purification, with JPEG compression and related transformations shown to reduce protection effectiveness (Sandoval-Segura, Geiping, & Goldstein, 2023; Xu & Singh, 2024; Guan et al., 2024; Zhao et al., 2024; Kim et al., 2025; Bala et al., 2025).
- Detection-based approaches remain fundamentally reactive: Li et al. (2023) report weak zero-shot generalization of deepfake-image detectors, while Yan et al. (2023) specifically address generalization through common forgery features, reinforcing the motivation for proactive protection.
- The reviewed protection methods show a continuing trade-off between protection strength, computational cost and image quality. Memory-efficient and learned approaches such as Lo et al. (2024), Guan et al. (2024), Qin et al.

(2024) and Liu et al. (2023) are therefore relevant to the project's edge-deployment objective.

- Provenance/authenticity mechanisms can complement adversarial protection, but the C2PA security analysis by Golaszewski et al. (2026) identifies weaknesses that mean provenance should be treated as a complementary verification layer rather than a standalone guarantee.
- Lightweight-architecture research such as MobileNetV4 (Qin et al., 2024) and EfficientViT (Liu et al., 2023) demonstrates practical routes toward lower memory and computation, supporting the feasibility of evaluating CogniShield on resource-constrained hardware.
- The reviewed literature spans diffusion-image immunization, facial-manipulation protection, style protection, compression-aware defenses, deepfake detection, provenance and efficient vision architectures. However, these components are generally evaluated separately rather than as one on-device pipeline combining proactive protection, robustness testing, visual-quality control and provenance.

Research Gap: Across the reviewed literature, adversarial image-protection methods, robustness mechanisms, efficient vision architectures and provenance/authenticity mechanisms are largely studied as separate components. The reviewed papers do not establish a single, lightweight pipeline that combines (a) specified target generative-manipulation models, (b) measurable visual-quality constraints, (c) compression/purification-aware robustness evaluation, (d) practical edge-oriented computation, and (e) complementary cryptographic provenance. CogniShield's Review-1 design is intended to investigate this combined gap.

## [7.] METHEDOLOGY:

The methodology of CogniShield follows a modular pipeline.

First, the input image is uploaded by the user. The system reads the image and converts it into RGB format. This ensures that all processing steps use a consistent image format.

Second, the image protection module generates a small perturbation. The perturbation is high-frequency noise, which is less visible to the human eye. The system also uses an edge-aware mask so that the perturbation blends better into natural image details. The perturbation is limited by an epsilon value, which controls the maximum pixel-level change.

Third, compression feedback is applied. Since online platforms compress uploaded images, the system simulates JPEG compression during protection. This helps make the protected image more practical for real-world conditions.

Fourth, the protected image is evaluated. The original and protected images are compared using PSNR, SSIM, MSE, mean absolute delta, and maximum absolute delta. These values are stored and displayed to the user.

Fifth, JPEG compression testing is performed at different quality levels. The protected image is saved or simulated at JPEG quality 95, 75, and 50. The system again calculates quality metrics after compression.

Sixth, cryptographic provenance is generated. The system calculates a SHA-256 hash of the protected image and signs it using Ed25519 private key cryptography. A provenance JSON file is generated for later verification.

Seventh, optional target model testing is performed using SD-Turbo. The original image and protected image are passed through the image-to-image model, and the outputs are visually compared.

## [8.] SOFTWARE REQUIREMENTS: Functional and Non-Functional Requirements

### Functional Requirements:

1. The system should allow the user to upload an image.
2. The system should generate a protected image using adversarial-style perturbation.
3. The system should display the original image, protected image, and difference image.
4. The system should calculate PSNR, SSIM, MSE, mean absolute delta, and maximum absolute delta.
5. The system should test the protected image under JPEG compression.
6. The system should generate a cryptographic provenance JSON file.
7. The system should verify whether the protected image has been modified after signing.
8. The system should allow the user to download the protected image and provenance file.

9. The system should optionally test the original and protected images using SD-Turbo.

**Non-Functional Requirements:**

1. The system should run locally on the user's device.
2. The system should preserve user privacy by avoiding cloud upload.
3. The protected image should remain visually similar to the original image.
4. The interface should be simple and easy to use.
5. The system should be modular so that new target models can be added later.
6. The system should support repeatable experiments using fixed seed values.
7. The system should generate measurable results suitable for project evaluation.

**Hardware Requirements:**

- Processor: Intel i5/Ryzen 5 or above recommended
- RAM: Minimum 8 GB, recommended 16 GB
- GPU: NVIDIA GPU recommended for SD-Turbo testing
- Storage: Minimum 20 GB free space for model files and outputs

**Software Requirements:**

- Operating System: Windows 10/11
- Python 3.12
- Streamlit
- NumPy
- Pillow
- OpenCV
- scikit-image
- cryptography
- PyTorch
- Hugging Face Diffusers
- Transformers
- SD-Turbo model files

## [9.] SYSTEM ARCHITECTURE:

The system starts with a user image. The image is uploaded through the Streamlit interface and preprocessed into RGB format. Then the adversarial protection module adds low-visibility perturbation using high-frequency noise, edge-aware masking, epsilon control, and compression feedback. The protected image is then evaluated using PSNR, SSIM, MSE, and pixel-difference metrics. It is also tested under JPEG compression. After that, the provenance module generates a SHA-256 hash and Ed25519 digital signature. Optionally, the original and protected images are tested against SD-Turbo to compare AI manipulation behavior. Finally, the system outputs the protected image, difference image, quality report, compression report, and provenance JSON.

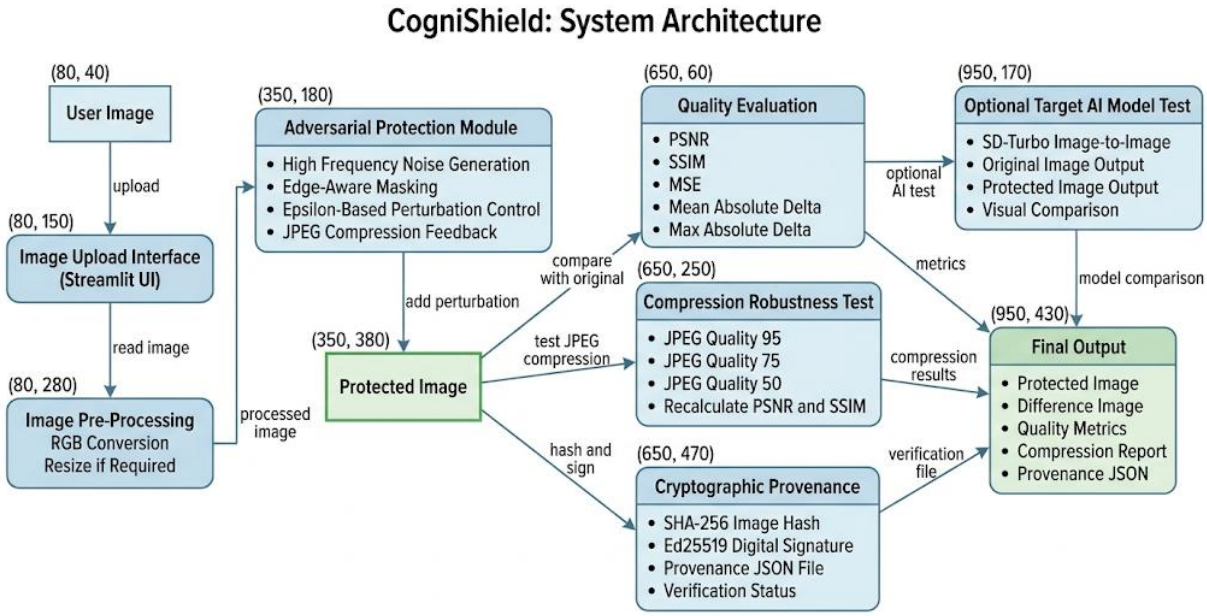



**Fig. 9.1** System Architecture of CogniShield

## [10.] UML DIAGRAMS:

### Use Case Diagram:

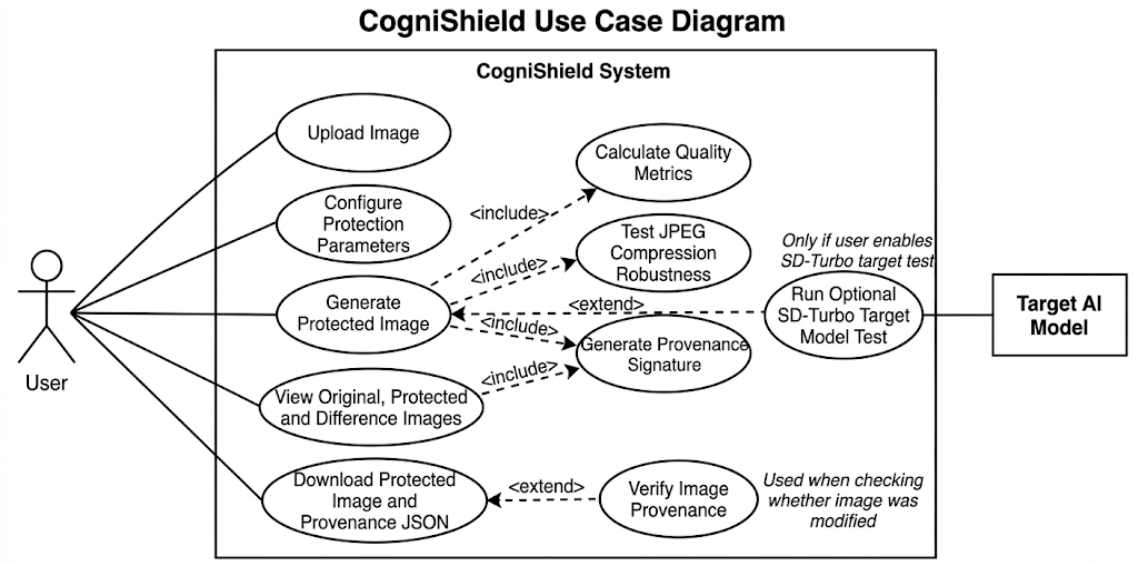



**Fig. 10.1** Use Case Diagram of CogniShield

### Class Diagram:

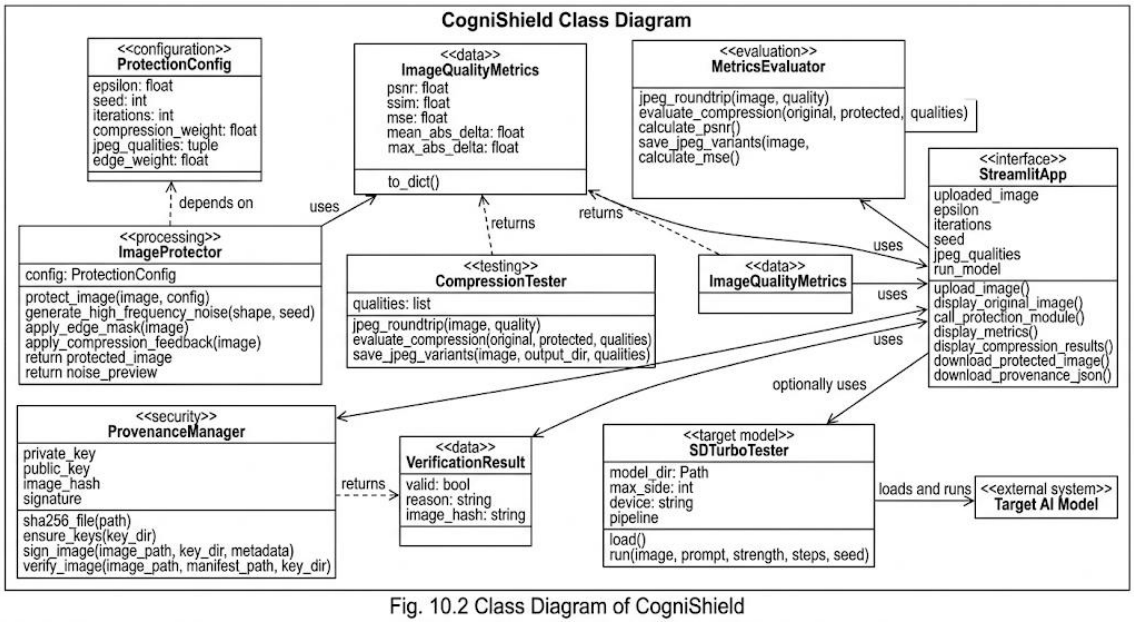



Fig. 10.2 Class Diagram of CogniShield

## [11.] SUMMARY:

CogniShield is a proactive digital image protection system designed to reduce misuse of personal images by generative AI tools. Unlike traditional fake media detection systems, CogniShield protects the image before manipulation occurs. The system adds low-visibility perturbations, evaluates visual quality, tests compression robustness, and generates cryptographic provenance.

The prototype successfully demonstrates image protection, PSNR and SSIM-based quality evaluation, JPEG compression testing, and image integrity verification using digital signatures. The system also supports optional SD-Turbo target model testing for comparing original and protected image behavior under generative AI transformation.

The current version is a working local prototype. Future improvements can include full gradient-based adversarial optimization, more target deepfake models, mobile deployment using ONNX or TensorFlow Lite, and social media upload/download testing. Overall, CogniShield provides a practical and privacy-preserving approach toward proactive digital identity protection.

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